Entry Information

ID:
AML.CS0012

Type:
Case-Study

Name:
Face Identification System Evasion via Physical Countermeasures

URL:
https://atlas.mitre.org/studies/AML.CS0012/

Incident Date:
2020-01-01

Target:
Commercial Face Identification Service

Actor:
MITRE AI Red Team

Description

MITRE's AI Red Team demonstrated a physical-domain evasion attack on a commercial face identification service with the intention of inducing a targeted misclassification.
This operation had a combination of traditional MITRE ATT&CK techniques such as finding valid accounts and executing code via an API - all interleaved with adversarial ML specific attacks.

Attack Procedure:
[AML.T0000] Search Open Technical Databases: The team first performed reconnaissance to gather information about the target ML model.
[AML.T0012] Valid Accounts: The team gained access to the commercial face identification service and its API through a valid account.
[AML.T0040] AI Model Inference API Access: The team accessed the inference API of the target model.
[AML.T0013] Discover AI Model Ontology: The team identified the list of identities targeted by the model by querying the target model's inference API.
[AML.T0002.000] Datasets: The team acquired representative open source data.
[AML.T0005] Create Proxy AI Model: The team developed a proxy model using the open source data.
[AML.T0043.000] White-Box Optimization: Using the proxy model, the red team optimized adversarial visual patterns as a physical domain patch-based attack using expectation over transformation.
[AML.T0008.003] Physical Countermeasures: The team printed the optimized patch.
[AML.T0041] Physical Environment Access: The team placed the countermeasure in the physical environment to cause issues in the face identification system.
[AML.T0015] Evade AI Model: The team successfully evaded the model using the physical countermeasure by causing targeted misclassifications.